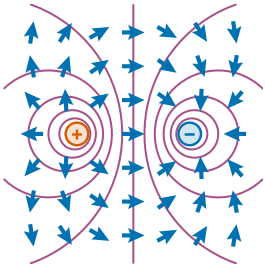
gradient type: field normal to contours

stream type: field along contours

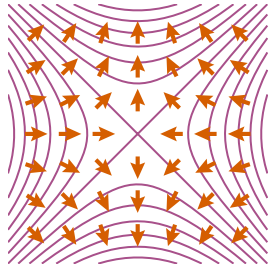
$\phi \rightarrow \mathbf{E} = -\nabla\phi$



electric potential · V

two opposite line charges

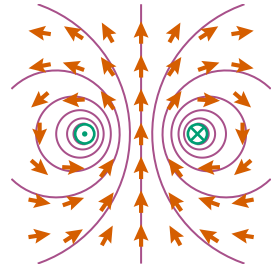
$\psi \rightarrow \mathbf{H} = -\nabla\psi$



magnetic scalar potential · A

current-free quadrupole

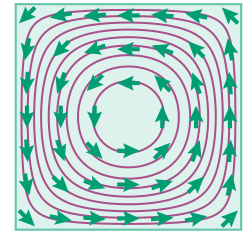
$A_z \rightarrow \mathbf{B} = \nabla A_z \times \mathbf{e}_z$



2-D vector potential · Wb/m

two antiparallel currents

$\psi_s \rightarrow \mathbf{K}_s = \nabla_s \psi_s \times \mathbf{n}$



surface stream function · A

current on a square plate